



Comparing Isolated Amplifiers And Isolated Modulators

This article covers the major differences between isolated amplifier-based and isolated modulator-based solutions, and explains their uses in improving the overall efficiency and power throughput in various industrial and automotive applications

Krunal Maniar
works as the business
development manager
at Texas Instruments,
Dallas, USA

Industrial applications such as motor drives, photovoltaic inverters, and uninterruptible power supplies (UPS), and automotive applications such as onboard chargers, traction inverters, and DC-DC converters operate at high voltage and current levels to optimise the overall efficiency and power throughput. These systems are subjected to hostile environments such as electrical noise, vibration, mechanical shock, extreme temperatures, ingress of contaminants, and so on.

Such systems demand robust, reliable, and galvanic isolation to isolate high voltages from low-voltage circuits. The feedback signals measured on these high voltages are galvanically isolated from the low-voltage controllers by isolated amplifiers or isolated modulators. This article compares isolated amplifier

and isolated modulator-based solutions and explains some unique advantages of isolated modulator-based solutions.

Introduction to isolated amplifiers

Fig. 1 shows the implementation of an isolated amplifier-based measurement solution.

The received bitstream and clock are synchronised and processed by an analogue low-pass filter on the low voltage side and presented as an analogue output signal. The differential output of the isolated amplifier is often converted to a single-ended analogue output with an op-amp-based circuit. This op-amp-based circuit can also implement a low-pass filter to further reduce the signal bandwidth to a bandwidth of interest and thereby improve the system's noise performance. The analogue-to-digital converter (ADC), either external or internal to the microcontroller (MCU) or digital signal processor (DSP), receives this feedback analogue output and converts this output back to the digital domain.

Introduction to isolated modulators

Fig. 2 shows the implementation of an isolated modulator-based measurement solution.

The input stage of an isolated modulator is similar to that of an isolated amplifier. The drivers transfer the modulator output across the isolation barrier. The isolated data output provides a digital

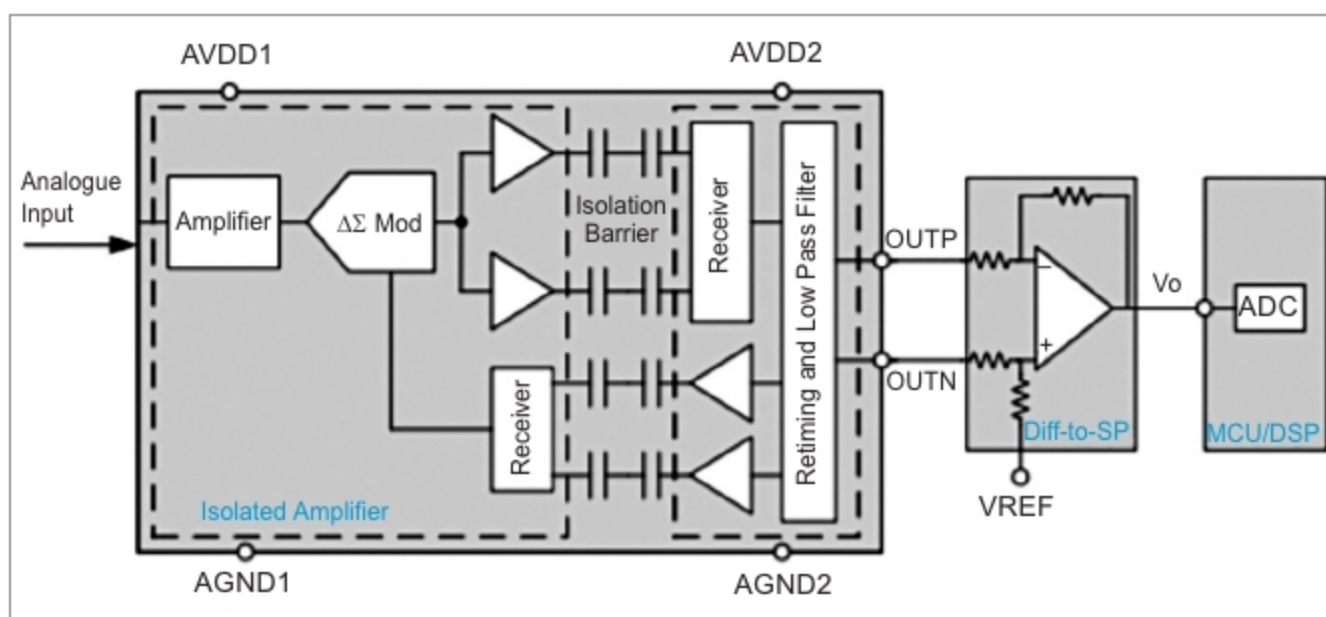


Fig. 1: Isolated amplifier implementation